

RHEA SUDHEER

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SUMMARY

Data scientist with an M.S. in Computer Science from Texas A&M and hands-on experience using Python, SQL, statistical analysis, predictive modeling, time-series forecasting, causal inference, and data visualization to turn large, multi-source datasets into actionable insights. Built analytical workflows across 100K+ public-policy records and 500K+ healthcare records, including data validation, feature engineering, model evaluation, metric design, dashboards, trend analysis, and stakeholder-facing reporting. Experienced translating ambiguous questions into measurable analyses and communicating model performance, trade-offs, and findings to technical and non-technical partners.

EDUCATION

Texas A&M University — Master of Computer Science, GPA: 3.83/4.00 | College Station, TX

Aug 2024 - May 2026

PES University — B.Tech. Computer Science, GPA: 9.1/10 | Bangalore, India

Dec 2020 - May 2024

TECHNICAL SKILLS

Programming & Querying: Python, SQL, R, C++, Java

Statistics & Analytics: Descriptive/Statistical Analysis, Correlation, Regression, Causal Inference, Cross-Validation, Model Evaluation, Exploratory Analysis

Modeling & Forecasting: Scikit-learn, XGBoost, Decision Trees, Ensemble Models, Predictive Modeling, Feature Engineering, ARIMA, Prophet, Time-Series Analysis

Data & Visualization: MySQL, Spark, ETL/Data Pipelines, Data Validation, Tableau, Dashboards, KPI/Metric Reporting

ML / Tools: PyTorch, TensorFlow, Keras, Git, AWS, GCP, Docker

EXPERIENCE

Public Policy Research Institute, Texas A&M University — Student Research Assistant

Jan 2025 - Present

- Analyzed and maintained multi-source public-policy datasets supporting 100,000+ records; used SQL to clean, join, aggregate, reconcile, and validate data for recurring analysis, dashboards, and reporting.
- Built API-backed ingestion, transformation, and automated data-quality workflows to identify malformed, incomplete, duplicated, and inconsistent records before downstream analysis, improving reliability of delivered metrics and datasets.
- Developed stakeholder-facing dashboards and recurring reports to surface trends, outcomes, and program-level metrics; translated evolving research questions into measurable definitions, data requirements, and analytical deliverables.
- Investigated anomalies and data-quality failures using SQL, logs, targeted validation, and root-cause analysis; partnered with researchers and analysts to communicate findings, clarify ambiguous requirements, and iterate on reporting logic.

IBM — Machine Learning Technical Mentee (3-month mentorship program)

Dec 2022 - Mar 2023

- Built an end-to-end Type-I diabetes prediction pipeline on heterogeneous multi-omic data using Python/Scikit-learn; performed preprocessing, EDA, feature engineering, cross-validation, and comparison of decision-tree and ensemble models, achieving 97.7% reported accuracy.
- Applied interpretability-oriented and causal-inference analysis to investigate influential variables; presented methodology, model-selection rationale, validation results, and implications to technical and non-technical audiences.

Ernst & Young — Technology Consulting Intern

Jan 2024 - Jun 2024

- Analyzed Oracle/SAP system evidence and operational data to identify access, change-management, data-integrity, and compliance gaps; performed root-cause analysis and communicated findings and remediation actions across client teams.

Thales — Artificial Intelligence Intern

Jun 2022 - Aug 2022

- Developed and evaluated a real-time MobileNetV2 + SSD pedestrian-detection pipeline; analyzed latency-vs.-accuracy trade-offs, false/missed detections, and deployment failure cases, and used SHAP/LIME/saliency methods to interpret model behavior.

PROJECTS

Healthcare Data Analysis, Readmission Prediction & Forecasting | Python, SQL, Scikit-learn, Statsmodels

Apr 2024

- Analyzed 500K+ hospital and insurance records using SQL + Python for feature extraction, exploratory analysis, trend discovery, and reusable analytical datasets; built regression and decision-tree models for readmission risk with 89% reported accuracy.
- Applied ARIMA and Prophet to forecast seasonal admission patterns and causal-inference analysis to assess the impact of care-policy changes on readmission outcomes, combining prediction, forecasting, and impact measurement.

Hybrid Self-RAG with External Knowledge Validation | Python, Sentence-Transformers, FAISS, LangChain

May 2025

- Designed a quantitative evaluation framework for an LLM retrieval system: decomposed generated answers into factual claims, matched claims to external evidence, embedded claim/evidence pairs, and defined a semantic FactScore from cosine similarity to measure grounding continuously rather than with a binary heuristic.
- Selected and evaluated a FactScore threshold of 0.5 to segment high- vs. low-confidence responses; low-scoring outputs triggered FAISS retrieval and evidence-conditioned regeneration, creating a measurable decision rule analogous to a production quality gate.
- Back-tested the hybrid pipeline against a Self-RAG-only baseline using factual accuracy, hallucination rate, average FactScore, external-validation behavior, and before/after regeneration analysis; factual accuracy improved 81% -> 89%, hallucination rate fell 19% -> 11%, and FactScore increased 0.65 -> 0.87.
- Performed example-level error analysis and quantified the quality/latency trade-off of stricter thresholds, live evidence APIs, embeddings, and conditional regeneration; used the analysis to identify threshold calibration, source quality, and retrieval failures as key levers for future performance.
- Validated threshold behavior on individual cases to understand when the quality gate should trigger: a low-grounding response improved from FactScore 0.21 to 0.525 after retrieval/regeneration, while a 0.69 response passed without unnecessary intervention.

Audio-Visual Speech Recognition with Transformers & CTC Loss | Python, PyTorch, Transformers, OpenCV

- Built and evaluated a multimodal speech-recognition pipeline using synchronized audio/video features, Transformer sequence modeling, and CTC loss; applied preprocessing, augmentation, and denoising and reduced recognition error rate by 5.7%.
- Performed robustness testing and error analysis across noisy-audio conditions, visual ambiguity, and multimodal alignment failures to identify which data and preprocessing changes drove measurable performance improvements.

PUBLICATIONS

[Revolutionizing Healthcare: A Review Unveiling the Transformative Power of Digital Twins](#) — IEEE Access, 2024

[Enhancing Robustness in Audio Visual Speech Recognition: A Preprocessing Approach with Transformer and CTC Loss](#) — AMATHE 2024

[Leukaemia: A Comparative Analysis of Deep Learning Models Using ALL Dataset](#) — RAICS 2024